Sem 1 Examination 2015

Question/answer booklet

YEAR 8
MATHEMATICS

Name: ______________________________

Teacher: ______________________________

Section Two:
Calculator-assumed

Time allowed for this section

Section One

Reading time before commencing work:	5 minutes
Working time for paper:	30 minutes

Material required/recommended for this paper

To be provided by the supervisor

This question/answer Booklet

Formula sheet

To be provided by the student

Standard items: pens, pencils, pencil sharpener, eraser, correction fluid/tape, ruler, highlighter

Special items: *drawing instruments, templates, notes on one unfolded sheet of A4 paper, and scientific calculator*

Important note to students

This test comprises of **TWO sections**. The **first section** is **calculator free** where no calculators of any kind are to be used. The **second section** is **calculator assumed** where a scientific calculator may be used. All questions must be answered in both sections. **Answers should be rounded appropriately**. All working should be shown in the space provided. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks.

No pen, pencils, highlights etc. may be used during reading time. This time is to be used to read through the assessment and check that you understand what is being asked of you. You may speak with the teacher/supervisor during this time (by putting up your hand and waiting patiently for them to approach you) but you may only ask clarification questions and not how to solve the problems. After reading time has ended, you may not ask any more questions.

6. [4 marks: 1, 1, 1, 1]

Complete the following conversions:

a) 1.25 km = 125000 cm **b)** 324589 mm = 324.589 m

c) 2 cm^2 = 200 mm^2 **d)** 1.35 m^2 = 13500 cm^2

7. [6 marks: 2, 2, 2]

Determine the area of the following:

a)

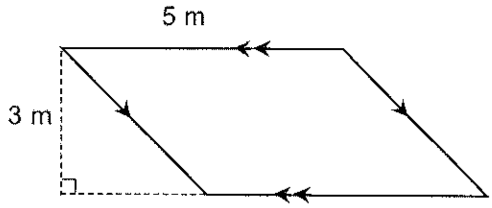
Rectangle 3.2 by 2.1cm

Area(rect) = L×W

= 3.2 × 2.1

= 6.72 cm^2

b)

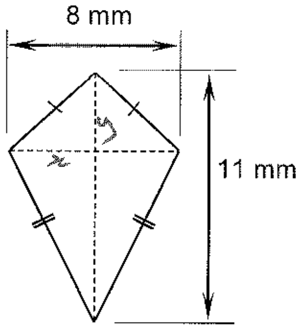


Area = ph × height

= 3m × 5m

= 15m^2

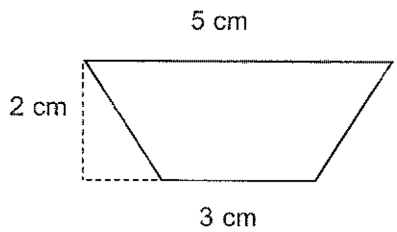
c)



Area(kite) = $\frac{1}{2} \times x \times y$

= $\frac{1}{2} \times 8 \times 11$

= 44 mm^2



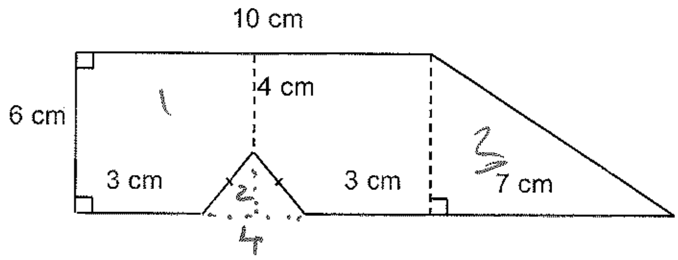
Area (Trapezoid) = $\frac{1}{2}(a+b) \times 2$

= $\frac{1}{2}(3+5) \times 2$

= 8 cm^2

8. [4 marks]

Determine the area of the following shape:



Area (Rectangle) = L × W

= 6 × 10

= 60 cm^2

Area (Little Δ) = $\frac{1}{2}$ b × ph

= $\frac{1}{2}$ × 4 × 2

= 4 cm^2

Area (Big Δ) = $\frac{1}{2}$ b × ph

= $\frac{1}{2}$. 7 . 6

= 21 cm^2

Total = Rectangle + Big Δ − Little Δ

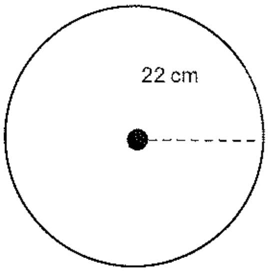
= 60 + 21 − 4

= 77 cm^2

9. [2 marks: 2]

Determine the perimeter of the following:

a)



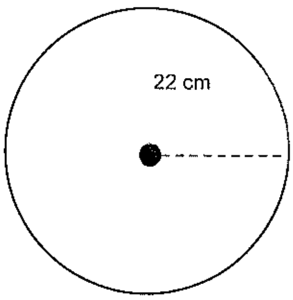
$C = 2\pi r$

$= 138.2\text{ cm}$

10. [4 marks: 2, 2]

Determine the area of the following:

a)

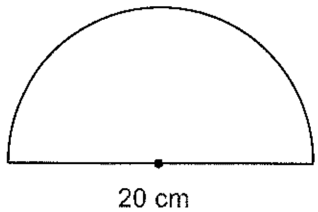


$A = \pi r^2$

$= \pi \times 22^2$

1520.5 cm^2

b)



$\text{Area} = \frac{1}{2}\pi r^2$

$= \frac{1}{2}\pi . 10^2$

$= 157.1\text{ cm}^2$

11 [4 marks]

Write down the following as recurring decimals

$\frac{1}{3}$ $0.\dot{3}$

$\frac{5}{6}$ $0.8\dot{3}$

12 [4 marks:1, 1,1,1] Simplify

$p + p + p + p =$ $4p$

$a + 2b + 3b + 2a =$ $3a+5b$

$x + 2xy =$ $x+2xy$

$7d - 3j + 2d + 2j =$ $9d-j$

13 [5 marks:1, 2, 2]

a

$p \times p \times p \times p =$ p^4

b If x = 5 what is the value of 20 -2x?

$20-2.5$
$20-10$
$=10$

c If 3x + 6 = 27 what is the value of x?

$3x+6=27$
$3x=21$
$x=7$

14 [4 marks:1, 1,1,1]
Match these words with the most accurate meaning.

Perimeter	Distance from one side of a circle to other going through the centre
Radius	The distance around the outside of a circle
Diameter	Distance around the outside of a shape
Circumference	Distance from the centre of a circle to it's edge

15. [2 marks]

If a rectangle has an area of $32cm^2$ give 2 possible sets of dimensions it could have.

4 x 8cm

1 mark per pair
max 2 marks

2cm x 16cm

1cm x 32cm

~ END OF SECTION TWO ~